

Math 421

Wednesday, October 22

Announcements

- Grace's Drop-In Hours
 - **Today** 12-1 pm VV 311
 - Thursday 9-11 am VV B205
- MLC
 - Proof Table: M-Th 3-7:30 VV B227
 - Course Assistant: Th 4-7 Table 4
- Homework 6 Due Friday

Bolzano-Weierstrass Theorem

$\{s_n: n \in \mathbb{N}\}$ is bounded (s_{n_k})
 $\{s_{n_k}: n_k \in \mathbb{N}\} \subseteq \{s_n: n \in \mathbb{N}\}$

Thm: Every bounded sequence has a convergent subsequence.

Discussion Question: Why is this theorem true?

11.4 - Every sequence has a monotonic subsequence.

10.2 - Every bounded monotonic sequence converges.

Ex: (r_n) is the sequence of rational numbers.
 $\mathbb{R} \cup \{-\infty, \infty\}$

Subsequential Limit

Def. Let (s_n) be a sequence in $\mathbb{R}$. A **subsequential limit** is any real number or symbol $+\infty$ or $-\infty$ that is the limit of some subsequence of (s_n) .

The set of subsequential limits is a subset of $\mathbb{R} \cup \{-\infty, +\infty\}$

Examples.

- (s_n) $s_n = (-1)^n$ $(-1, 1, -1, 1, \dots)$ $\{-1, 1\}$
 - $(-1, -1, -1, \dots)$ converges to -1
 - $(1, 1, 1, \dots)$ converges to 1
- $(t_n) = (0, 1, 0, 2, 0, 3, \dots)$ $\{0, +\infty\}$
 - $(0, 0, 0, 0, \dots)$ converges to 0
 - $(1, 2, 3, \dots)$ diverges to $+\infty$

$$(z_n) = (0, 1, -1, 2, -2, \dots) \quad \{-\infty, \infty\}$$

Activity

For each of the following sequences, determine its set of subsequential limits.

1. (s_n) where $s_n = \frac{1}{n}$. $\{0\}$
2. (a_n) where $a_n = n^2(-1)^n$. $\{-\infty, \infty\}$
3. (b_n) where $b_n = \sin\left(\frac{n\pi}{3}\right)$. $\{-\frac{\sqrt{3}}{2}, 0, \frac{\sqrt{3}}{2}\}$
4. (r_n) a list of all rational numbers. $\mathbb{R} \cup \{-\infty, \infty\}$
5. (c_n) where $c_n = n(1 + (-1)^n)$. $\{0, +\infty\}$

Chapter 2: Sequences

Section 10 (Part 2), Section 11 (Part 2) & Section 12: $\limsup$ and $\liminf$

$$S_1 = \left\{ \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots \right\} \quad S_5 = \left\{ \frac{1}{6}, \frac{1}{7}, \frac{1}{8}, \dots \right\} \quad S_{10} = \left\{ \frac{1}{11}, \frac{1}{12}, \frac{1}{13}, \dots \right\}$$

Activity

$$S_1 \supseteq S_5 \supseteq S_{10}$$

Consider the sequence defined by $s_n = \frac{1}{n}$.

$$\left[t_n = -\frac{1}{n}, a_n = (-1)^n \frac{1}{n} \right]$$

- Define the sets $S_N = \{s_n : n > N\}$, the “tails” of the sequence. Write out a couple of these sets, $N = 1, 5, 10$ for example.
- What is the relationship between these sets?
 $S_{n_1} \supseteq S_{n_2}$ for $n_1 > n_2$
 $S_{n_1} \subseteq S_{n_2}$ for $n_1 > n_2$
- Define $u_N = \inf\{s_n : n > N\}$ and $v_N = \sup\{s_n : n > N\}$. Find these values for the sets you looked at in 1.
- What do you notice about the relationship between the u_N and the v_N ?
 $u_1 = u_5 = u_{10} = 0$
 $v_1 > v_5 > v_{10}$
 $(=, \leq, \geq, \text{etc.})$
- What is $u = \lim_{N \rightarrow \infty} u_N$? $v = \lim_{N \rightarrow \infty} v_N$?
 $u = 0$
 $v = 0$

The sets $\{s_n : n > N\} = S_N$

- $s_1 \geq s_2 \geq s_3 \geq \dots$

- Define $u_N = \inf \{s_n : n > N\}$ $v_N = \sup \{s_n : n > N\}$ we get sequences
 (u_N) and (v_N) such that

- $u_1 \leq u_2 \leq u_3 \leq \dots$ and $v_1 \geq v_2 \geq v_3 \geq \dots$

Monotonic so limit exists.

- Define $u = \lim_{N \rightarrow \infty} u_N$ $v = \lim_{N \rightarrow \infty} v_N$.